Overall framework for ultrasound bone classification

max1 + channel4 / full_dynamic_envelope / 6ViT + cross-vote

A | Shared-frame selection

Raw ultrasound
50 × 2 × 896

ADC zero-level correction

Dynamic locator
All 50 frames
Both beams

Beam 1: Top10 peak |amplitude|
Beam 2: Top10 peak |amplitude|

Intersect the two frame-ID sets
Select max Beam 1 peak in overlap

Separate main windows per beam; boundaries are reused.
Empty overlap → legacy similarity fallback within Beam 1 Top10
Same selected frame index for both beams and all three regions

B | Dynamic regions and preprocessing

full_dynamic_envelope (code: dyn_envelope_full)

Main window
170 samples

Continuous tail
600 samples

Full signal
896 samples

Tail starts at main-window end; right-pad beyond signal end.

Per region × beam: cubic spline → 512 samples → Tukey (α = 0.3)
Signed waveforms retained • Model input: [N, 6, 1, 512]

C | Six independent 1D ViT branches

Each branch has its own encoder and K-class head

Main · Beam 1	1D ViT encoder 1	Head 1 + softmax	p1(c)
Main · Beam 2	1D ViT encoder 2	Head 2 + softmax	p2(c)
Tail · Beam 1	1D ViT encoder 3	Head 3 + softmax	p3(c)
Tail · Beam 2	1D ViT encoder 4	Head 4 + softmax	p4(c)
Full · Beam 1	1D ViT encoder 5	Head 5 + softmax	p5(c)
Full · Beam 2	1D ViT encoder 6	Head 6 + softmax	p6(c)

Encoder: Conv1D patches + fixed position → Transformer → mean pool → LayerNorm

D | Cross-vote

Sum class probabilities
$S(c) = \Sigma\ p_b(c),\ b = 1\ldots6$
$\hat{y} = \text{argmax}\ S(c)$

Predicted bone class

Equal-weight soft voting

Training: −log mean branch probability of the true class • Early stopping on validation loss • Inference: argmax